

Improving the Resolution of Chromosome Imaging by High Numerical Aperture Fourier Ptychography Microscopy

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ABSTRACT

Metaphase chromosome karyotyping plays an important role in the diagnosis of certain cancers and some genetic diseases by detecting chromosome abnormalities. For this technique, high magnification objective lens is used to ensure the chromosome's band pattern sharpness, but the small field of view (FOV) of the lens makes the imaging of chromosomes very tedious and time consuming. The purpose of this study is to verify the use of the Fourier ptychography microscopy (FPM) system in high-resolution karyotyping. Based on our former study, we further expanded the theoretical NA of the FPM system to 1.11 with a 20×/0.4 NA objective lens and higher illumination angles. To evaluate the resolving power of the FPM system, a 1951 USAF resolution target was imaged to create the modulation transfer function (MTF) curves. The performance of the FPM system was also assessed by imaging chromosomes acquired from blood and bone marrow pathological samples. The results were compared with a conventional 100×/1.45 NA oil immersion objective lens. The MTF curves demonstrate that the contrast of the FPM system is inferior but close to the 100× objective lens (1.45 NA). As compared to the images acquired by the 100×/1.45 NA oil immersion objective lens, the chromosome images recovered by the FPM system contain all the band patterns, despite the loss of some fine details. This study initially verified that the high NA FPM system can guarantee the sharpness of chromosome band patterns as the conventional high magnification oil immersion objective lens, while enabling a large FOV without the utilization of oil immersion medium.

Keywords: Fourier ptychography microscopy, chromosome karyotyping, high throughput scanning.

1. INTRODUCTION

Being a critical part of chromosome karyotyping to diagnose certain genetic diseases and cancers¹, locating and imaging the analyzable metaphase chromosome on whole pathological slides is very laborious and time consuming. Therefore, many studies have been focused on developing automated computer-assisted karyotyping schemes to reduce the clinicians' workload, by trying to identify and classify metaphase chromosomes automatically^{2,3}. However, the small field of view (FOV) of high magnification objective lens required by the karyotyping results in a long scanning time, which hampers their performance in efficient chromosome karyotyping.

The recently invented Fourier ptychography microscopy (FPM) provides a solution to effectively expand the space-bandwidth product of microscopes through computation^{4,5}. Instead of using high magnification objective lens, FPM first captures a series of low-resolution and large FOV images when the sample is illuminated by plane waves from varying angles. Then the obtained images are stitched together iteratively in Fourier domain to recover a high-resolution complex

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image (i.e., intensity and phase) by a phase retrieval algorithm. The feasibility of FPM's application in the clinical imaging of metaphase chromosomes in blood samples has been discussed and verified, in which 4×/0.1 NA and 10×/0.25 NA objective lenses were used to achieve equivalent NA_{eq} of ~0.48 and ~0.63, respectively⁶. The performance of FPM equipped with 10×/0.25 NA objective lens is better than conventional 60×/0.95 NA objective lens and is close to 100×/1.25 NA objective lens. Furthermore, since the low magnification objective is used, FPM system shows both desirable larger FOV and large depth of field (DOF) in chromosome imaging.

In this study, we further enhanced the equivalent NA of the FPM system by equipping a 20×/0.4 NA objective lens and decreasing the LED-sample distance. The corresponding modulation transfer function (MTF) curves were utilized to evaluate its resolution. Then we obtained images of chromosomes in blood and bone marrow samples of leukemia patients using the FPM system, and compared them with the images acquired by a conventional high magnification microscope equipped with a 100×/1.45 NA oil immersion objective lens. The experimental set-up and imaging results are briefly presented in the following parts.

2. MATERIALS AND METHODS

The FPM-based microscope setup used in this experiment was built with components of Thorlabs Cerna Microscope platform (Thorlabs, New Jersey, USA), one FL20BW CMOS camera (Tuscan, Fuzhou, China), and one 32×32 LED panel (Adafruit, New York, USA). A 20×/0.4 NA Olympus objective lens (Olympus, Tokyo, Japan) was equipped on this microscope with a 150mm tube lens, resulting in a total magnification of approximately 16.4×. The LED array was placed 47mm under the sample stage to generate green incident light (central wavelength 0.53μm) from different angles. During the image collection, the 49 LEDs inside the central circular area with a radius of 48mm on the LED array were lit up in sequence to provide an equivalent system NA_{eq} of 1.11. As a comparison, we captured high magnification images under a conventional Nikon Ni transmissive microscope (Nikon, Tokyo, Japan) equipped with a 100×/1.45 NA oil immersion objective lens.

FPM is a computational imaging method combining aperture synthesis⁷ and phase retrieval technique⁸. In FPM setup, the objective lens is regarded as a low-pass filter⁹ with a radius defined by its NA_{obj} and wavelength in vacuum λ : $2\pi NA_{obj}/\lambda$. The illumination plane wave with specific oblique incident angle θ shifts the Fourier spectrum of the sample's complex optical field by $2\pi \sin\theta/\lambda$ in the back focal plane of the objective lens, and a subregion on the shifted spectrum is then collected by the objective lens to generate a low-resolution image containing intensity information. If the illumination angles are configured properly, the subregions captured under different illumination angles will overlap and cover a larger area of the sample's Fourier spectrum. Then the phase retrieval technique will restore the high-resolution amplitude and phase information iteratively, with the captured intensity images as constraints^{9, 10}. Thus, the NA_{eq} of the system is determined by the NA of the objective and maximum illumination angle θ_{max} , and it can be calculated by¹¹: $NA_{eq} = NA_{obj} + NA_{ill}$, where NA_{ill} denotes the illumination NA and equals to $\sin\theta_{max}$.

Next, we generated the MTF curve of the FPM system using a USAF 1951 resolution target and compared it with the conventional microscope. The measured maximal pixel value I_{max} and minimal pixel values I_{min} along bar patterns of discrete spatial frequencies were used to evaluate the contrast values by^{12, 13}: $C = (I_{max} - I_{min})/(I_{max} + I_{min})$. After normalization a smooth MTF curve was generated by a curve fitting algorithm.

Similarly, the analyzable metaphase chromosomes in standardly prepared blood and bone marrow clinical samples were imaged under both FPM (equipped with 20×/0.4 NA objective lens) and conventional microscope systems (equipped with 100×/1.45 oil immersion NA objective lens). The sharpness of chromosome band patterns on obtained images was used to compare the FPM system with conventional microscope system. Since phase shift on the background area of the clinical samples is close to zero, which results in a dark background on the raw phase images, we particularly inverted the pixel values when displaying phase images.

3. RESULTS

The high-resolution images of a USAF resolution target recovered by FPM are shown in Figure 1. The original image captured under the 20×/0.4 NA objective lens with conventional illumination is illustrated in Figure 1a. The resolvable patterns reach up to group 9-5 (812.7 lp/mm), larger than the theoretical result (618.6 lp/mm). Figure 1b demonstrates the corresponding FPM recovered image. The spatial resolution of the system is enhanced up to 2048 lp/mm (Group 11-1), which is inferior but close to the image captured under the conventional microscope's 100×/1.45 NA oil immersion objective lens (Figure 1c). Accordingly, Figure 2 shows the measured MTF curves of the FPM-based and

conventional microscopes. The MTF curve of 100 $\times$ /1.45 NA oil immersion objective lens in Figure 2b has higher contrast value than the FPM system shown in Figure 2a, resulting in a higher spatial resolution of 1502.5 lp/mm at a contrast of 0.7 and a higher cutting off frequency of 2640.2 lp/mm than those of the FPM system (1298.3 lp/mm and 2373.1 lp/mm, respectively).

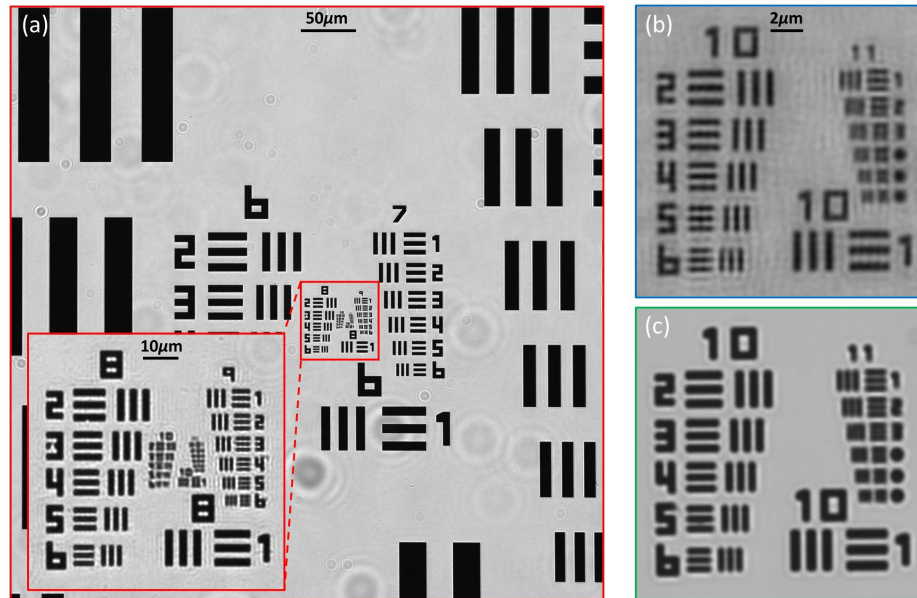


Figure 1. The microscopic images of a standard resolution target (USAF 1951), acquired via a) 20 $\times$ /0.4 NA objective lens with conventional illumination; b) FPM equipped with 20 $\times$ /0.4 NA objective lens; c) conventional microscope equipped with 100 $\times$ /1.45 NA oil immersion objective lens.

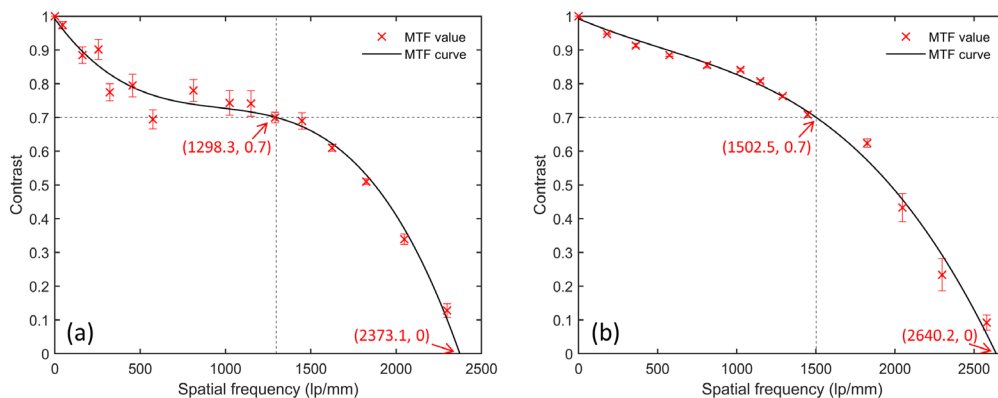


Figure 2. The MTF curves of a) FPM technology equipped with 20 $\times$ /0.4 NA objective lens and b) conventional microscope equipped with 100 $\times$ /1.45 NA oil immersion objective lens.

Next, one metaphase chromosome in a pathological sample of blood was imaged by the FPM system and conventional microscope, as shown in Figure 3. The chromosome image captured directly under the 20 $\times$ /0.4 NA objective lens (Figure 3a) has recognizable band patterns, but it is noticeable that the detailed band patterns are blurred and unresolvable when compared with the image captured under 100 $\times$ /1.45 NA oil immersion objective lens (Figure 3b). As expected, the FPM technique (under 20 $\times$ /0.4 NA objective lens) considerably improves the image resolution of the

chromosome in the blood sample (Figure 3c1 and c2). In the recovered amplitude image (Figure 3c1), the quality of bands is very close to the result of 100 $\times$ /1.45 NA oil immersion objective lens (Figure 3b), despite the loss of some fine details. Similarly, the FPM system clearly resolves chromosome bands in a pathological bone marrow sample, as shown in Figure 4. The image quality of recovered amplitude (Figure 4c1) is significantly better than the raw image captured directly under 20 $\times$ /0.4 NA objective lens (Figure 4a) and close to the performance of the high magnification 100 $\times$ /1.45 NA oil immersion objective lens (Figure 4b).

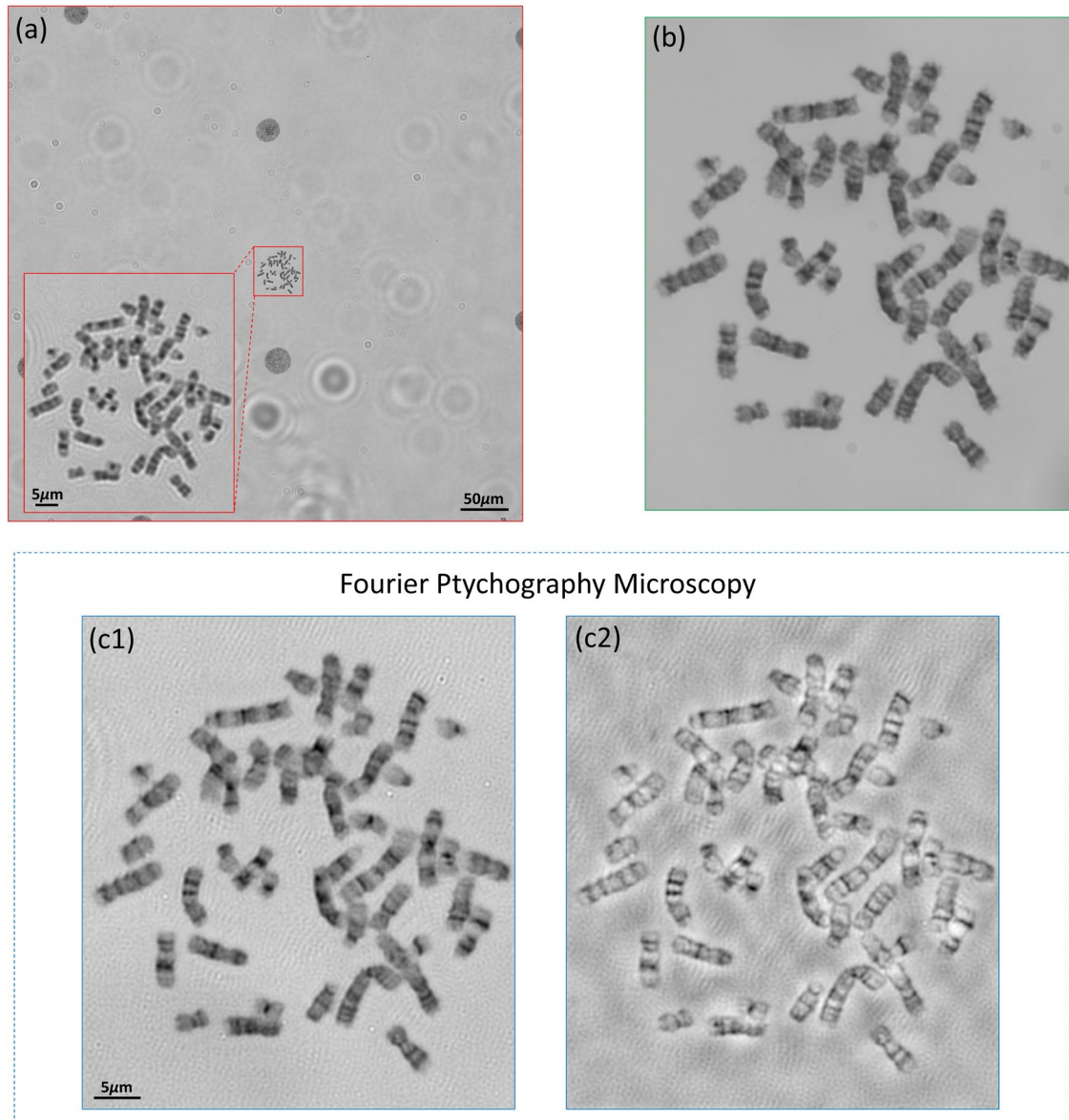


Figure 3. The sample images of a chromosome in the blood sample of one leukemia patient, obtained using a) 20 $\times$ /0.4 NA objective lens with conventional illumination; b) conventional microscope equipped with 100 $\times$ /1.45 NA oil immersion objective lens; c1, c2) FPM with 20 $\times$ /0.4 NA objective lens. c1) and c2) are recovered amplitude and phase results, respectively.

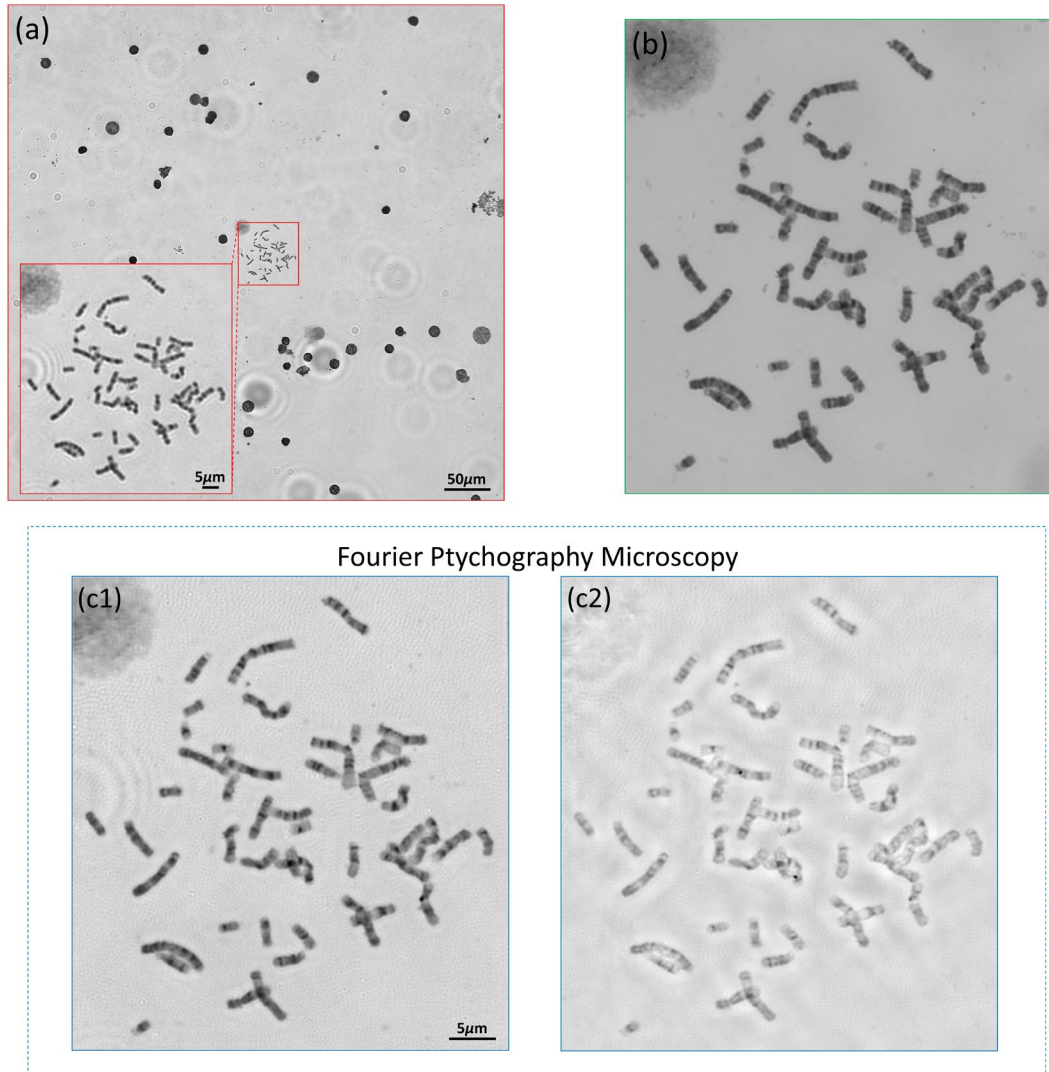


Figure 4. The sample images of a chromosome in the bone marrow sample of leukemia patients, obtained using a) 20 $\times$ /0.4 NA objective lens with conventional illumination; b) conventional microscope equipped with 100 $\times$ /1.45 NA oil immersion objective lens; c1, c2) FPM with 20 $\times$ /0.4 NA objective lens. c1) and c2) are recovered amplitude and phase results, respectively.

4. DISCUSSION AND CONCLUSION

Compared to the previous study⁶, we further enhanced the NA of the FPM system by using higher NA objective lens and larger incident angles in illumination and verified its application in chromosome karyotyping. The FPM system with large FOV and high spatial resolution can be realized in low cost, which is achieved by employing a simple LED matrix on a conventional microscope. Meanwhile, the FPM technology can also avoid the use of an oil immersion medium on the samples. The smallest pattern on the reconstructed USAF resolution target image is group 11-1 (Figure 1b), corresponding to a system NA of ~ 1.32 which is larger than the theoretical system NA of 1.11. This can be partially attributed to the fact that the actual NA of the 20 $\times$ /0.4 NA objective lens is ~ 0.53 (corresponding to group 9-5 in Figure 1a). The FPM reconstructed results of chromosomes in both blood and bone marrow samples have close quality to the conventional microscope under 100 $\times$ /1.45 NA oil immersion objective lens. Thus, the high NA FPM system can be used to ensure the quality of the band patterns for chromosome karyotyping.

Despite the motivating results, there are still several ways to improve the performance of the high NA FPM system. First, a condenser can be placed on top of the LED matrix without changing any other parts of current FPM setup, to improve the light collection efficiency of the system^{14, 15}. Second, to increase the intensity and signal to noise ratio under high illumination angles, the planer LED matrix in the current setup can be replaced by a hemispherical digital condenser¹⁶. This also allows us to use larger illumination angles during image acquisition. Some system corrections methods^{17, 18} might also enable us to further improve the reconstruction results.

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REFERENCES

- [1] J. Piper, [Automated Cytogenetics in the Study of Mutagenesis and Cancer] Springer Berlin Heidelberg, Berlin, Heidelberg, (1990).
- [2] F. Abid, and L. Hamami, "A survey of neural network based automated systems for human chromosome classification," *Artificial Intelligence Review*, 49(1), 41-56 (2018).
- [3] X. Wang, B. Zheng, S. Li, *et al.*, "Development and assessment of an integrated computer-aided detection scheme for digital microscopic images of metaphase chromosomes," *Journal of Electronic Imaging*, 17(4), 043008 (2008).
- [4] G. Zheng, R. Horstmeyer, and C. Yang, "Wide-field, high-resolution Fourier ptychographic microscopy," *Nature Photonics*, 7(9), 739-745 (2013).
- [5] G. Zheng, C. Shen, S. Jiang, *et al.*, "Concept, implementations and applications of Fourier ptychography," *Nature Reviews Physics*, 3(3), 207-223 (2021).
- [6] K. Zhang, X. Lu, X. Chen, *et al.*, "Using Fourier Ptychography Microscopy to Achieve High Resolution Chromosome Imaging: An Initial Evaluation," *Journal of Biomedical Optics*, accepted pending minor revision.
- [7] S. A. Alexandrov, T. R. Hillman, T. Gutzler, *et al.*, "Synthetic Aperture Fourier Holographic Optical Microscopy," *Physical Review Letters*, 97(16), 168102 (2006).
- [8] J. R. Fienup, "Phase retrieval algorithms: a comparison," *Applied Optics*, 21(15), 2758-2769 (1982).
- [9] X. Ou, G. Zheng, and C. Yang, "Embedded pupil function recovery for Fourier ptychographic microscopy," *Optics Express*, 22(5), 4960-4972 (2014).
- [10] L. Tian, X. Li, K. Ramchandran, *et al.*, "Multiplexed coded illumination for Fourier Ptychography with an LED array microscope," *Biomedical Optics Express*, 5(7), 2376-2389 (2014).
- [11] X. Ou, R. Horstmeyer, G. Zheng, *et al.*, "High numerical aperture Fourier ptychography: principle, implementation and characterization," *Optics Express*, 23(3), 3472-3491 (2015).
- [12] Y. Qiu, X. Chen, Y. Li, *et al.*, "Impact of the optical depth of field on cytogenetic image quality," *J Biomed Opt*, 17(9), 96017-1 (2012).
- [13] W. J. Smith, [Modern Optical Engineering], 4th ed, McGraw-Hill, New York, (2008).
- [14] G. Zheng, C. Kolner, and C. Yang, "Microscopy refocusing and dark-field imaging by using a simple LED array," *Optics Letters*, 36(20), 3987-3989 (2011).
- [15] J. Sun, C. Zuo, L. Zhang, *et al.*, "Resolution-enhanced Fourier ptychographic microscopy based on high-numerical-aperture illuminations," *Scientific Reports*, 7(1), 1187 (2017).
- [16] A. Pan, Y. Zhang, K. Wen, *et al.*, "Subwavelength resolution Fourier ptychography with hemispherical digital condensers," *Optics Express*, 26(18), 23119-23131 (2018).
- [17] A. Pan, Y. Zhang, T. Zhao, *et al.*, "System calibration method for Fourier ptychographic microscopy," *Journal of Biomedical Optics*, 22(9), 096005 (2017).
- [18] J. Sun, Q. Chen, Y. Zhang, *et al.*, "Efficient positional misalignment correction method for Fourier ptychographic microscopy," *Biomedical Optics Express*, 7(4), 1336-1350 (2016).